

Mason Deal

Raleigh, North Carolina

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Objective: Computer engineering senior seeking a Summer 2027 internship combining hands-on engineering, debugging, customer/product interaction, and system-level understanding in computer hardware engineering.

Education:

North Carolina State University | Raleigh, North Carolina

GPA: 3.61 | B.S. expected May 2027

Bachelor of Science (B.S.) in Computer Engineering

Accelerated Bachelor's-Master's (ABM) track with graduate coursework in progress

Technical Skills:

Programming Languages & Software: C/C++, RTL Verilog, Java, Assembly, Git

Hardware: MSP/Arm Embedded Systems, PCB Design, FPGA/ASIC Design, Microprocessor Architecture

Tools & Simulation: AMD Vivado, Altium, SPICE, Oscilloscopes, Logic Analyzers

Operating Systems: Windows, macOS, Linux (Ubuntu and Debian)

Work Experience:

Texas Instruments | Dallas, Texas

May 2026 - August 2026

Applications Engineer Intern (Battery Gauge Products)

- Collaborated cross-functionally with industrial and consumer electronics customers to diagnose, debug, and resolve application-level hardware failures and integration issues.
- Developed an embedded Linux kernel software driver suite to enable system compatibility and I2C serial communications for a specific TI battery gauge IC architecture.
- Designed and validated a custom PCB via Altium, incorporating electrostatic discharge (ESD) protection architectures to guarantee reliable gauge data integrity when encountering ESD events.

North Carolina State University Libraries | Raleigh, North Carolina

January 2025 - Present

Student IT Technician

- Maintain and troubleshoot 1,000+ Windows, macOS, and Linux systems by diagnosing hardware, software, and network issues, improving device reliability, and minimizing downtime across campus workstations.
- Resolve complex system failures through configuration analysis, system imaging, and documented troubleshooting procedures, preventing recurring technical issues and improving support efficiency.
- Deliver responsive technical support in a customer-facing helpdesk environment, strengthening skills in hardware diagnostics, system deployment, and technical communication.

Superior Concrete of NC Inc. | Greenville, North Carolina (hybrid)

February 2021 - Present

Assistant Estimator

- Designed and implemented a digital project archival system to organize historical project records, significantly improving internal data retrieval and workflow efficiency.
- Expanded company server storage infrastructure by 5 TB, enabling reliable long-term data retention.
- Launched a redesigned company website and resolved software, server, and network access issues through systematic technical troubleshooting and systems analysis.

Leadership & Organizations:

Alpha Phi Omega National Service Fraternity

August 2024 - Present

- **Education Vice President**, previous Webmaster, External Shadow, Scouting Fellowship chairs.

Institute of Electrical and Electronics Engineers (IEEE) Student Branch

January 2025 - Present

- **Treasurer**, previous Web Development Lead: won 1st place at SouthEastCon 2026 for best website.

Apple Next-Gen Innovators Mentorship - Mentee

September 2025 - Present

Honors & Awards:

Eagle Scout, University Honors Program, Dean's List (Spring 2025, Spring 2026), 2026 NC State University Libraries Student Assistant of the Year, TI Engineer Your Future Participant, SECU "People Helping People" Scholarship

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Automated Datacenter Liquid Cooling System Testing (in progress)

Lenovo / NC State - *Multidisciplinary Senior Design*



Lenovo public product imagery

Description:

Developing an automated testing system for Neptune liquid cooling systems used in Lenovo enterprise/datacenter server racks in collaboration with Lenovo's manufacturing organization. I am a Technical Project Manager responsible for coordinating system engineering efforts across electrical, mechanical, and industrial engineering disciplines.

Engineering Focus:

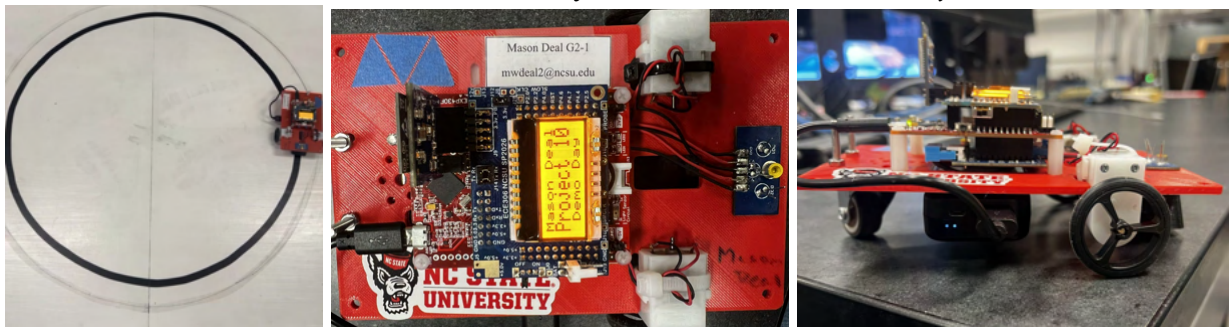
- System integration and test automation
- Hardware/software interfaces
- Cross-disciplinary engineering

Project Status:

Currently in the requirements and system-design phase. Technical details and project results will be added as development progresses.

Embedded Autonomous & WiFi Remote Controlled Car

North Carolina State University - *Introduction to Embedded Systems*



Description:

Engineered a dual-mode autonomous and WiFi-enabled vehicle utilizing the Texas Instruments MSP430FR2355 microcontroller. Redesigned the power delivery system by migrating from AA batteries to a rechargeable lithium-ion solution, significantly enhancing system stability and operational uptime. Through testing and adaptive PID control algorithms, the vehicle completed a combined autonomous and Wi-Fi-controlled course on its first attempt.

Challenges faced:

- Developed low-level firmware to manage peripheral integration, including hardware timers, interrupts, and serial IoT communication protocols.
- Managed complex hardware-software integration by interpreting schematics, debugging electrical systems, and working extensively with component datasheets.

How would I improve on this?

- Explore DAC-based voltage control for more precise speed regulation.
- Implement software-defined low-power sleep modes to improve energy efficiency and extend battery life.

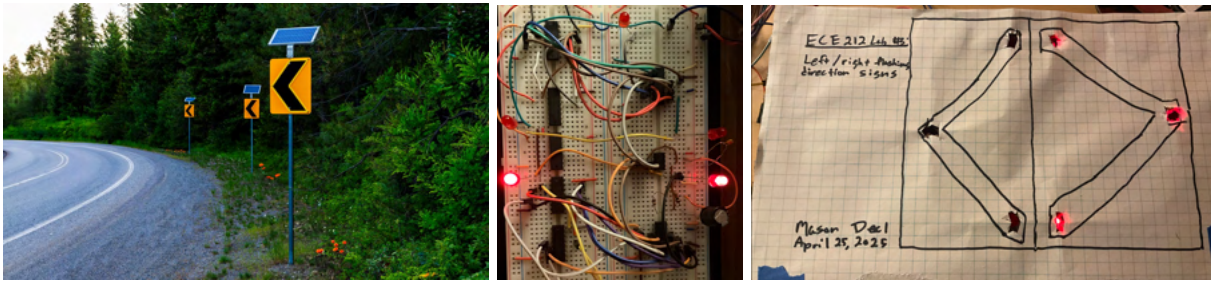
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LED Road Sign CMOS State Machine

North Carolina State University - *Introduction to Logic Design*



Description:

Designed and breadboarded a sequential logic circuit to emulate a curve-warning road sign, adhering to the full engineering design lifecycle from schematic design to physical demonstration.

Challenges faced:

- Designing and constructing two concurrent Finite State Machines (FSMs) to handle asynchronous user inputs and multi-pattern flashing sequences.
- Synchronized complex logic timing by troubleshooting and stabilizing the integrated clock circuit.

How would I improve on this?

- Integrate additional logic states to support synchronized "hazard" flashing modes.
- Optimize physical layout and wire routing to minimize interference and improve long-term reliability.